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Androgenetic Alopecia

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Continuing Education Activity

Androgenetic **alopecia** is a genetically predetermined disorder due to an excessive response to androgens. This condition affects up to 50 percent of males and females and is characterized by progressive loss of terminal hair of the scalp any time after puberty. It follows a characteristic distribution in both males and females. Hair loss is most prominent in the vertex and frontotemporal regions in males. In women, the frontal hairline is typically spared with diffuse hair loss at the crown and top of the head, with loss often marked by a wider center part. This activity examines when this condition should be considered in differential diagnosis, steps for evaluation, and treatment options. This activity highlights the role of the interprofessional team in caring for patients with this condition using different modalities tailored to individual needs and presentations.

Objectives:

- Identify the clinical manifestations and patterns of androgenetic **alopecia** in patients of different genders and ages.
- Screen for cardiovascular risk factors, hormonal imbalances, and other potential comorbidities associated with androgenetic **alopecia**.
- Implement evidence-based treatment strategies based on individual patient characteristics and preferences, including medications, low-level laser therapy, or hair transplantation.
- Coordinate comprehensive care plans for patients, integrating medical and psychological support to enhance the overall management of androgenetic **alopecia**.

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Introduction

Pattern or androgenetic **alopecia** is a genetically predetermined disorder due to excessive response to androgens, impacting approximately 50% of males and females.[1] This condition is characterized by progressive loss of terminal hair of the scalp, typically occurring following puberty, with a distinctive pattern in both males and females. In males, hair loss is most prominent in the vertex and frontotemporal regions, whereas females tend to retain their frontal hairline while experiencing diffuse apical hair thinning, causing a broader appearance of the anterior part of the hair.[2][3][4]

Etiology

Androgenetic **alopecia**, as indicated by its name, possesses a distinct genetic predisposition and is predominantly due to an excessive response to androgen.[5][6] This condition is characterized by a polygenic nature with varying penetrance degrees, influenced by maternal and paternal genes. A familial predisposition towards androgenetic **alopecia** exists, with sons having 5 to 6 times higher relative risk if their fathers experienced balding.

The onset of pattern **alopecia** is contingent upon androgen receptor activation and is typically observed after puberty. Prepubertal castration and individuals with androgen insensitivity syndrome do not exhibit pattern baldness. Both hormone metabolism and androgen receptors play a key role in developing pattern **alopecia**.[7]

Epidemiology

Caucasian individuals are most notably more affected, followed by Asians and African Americans, and subsequently by Native American and Inuit populations. The incidence aligns closely with age in Caucasian males, as approximately 50% are affected by 50 and approximately 80% by age 70. The disorder is quite common in females, with its incidence showing a notable rise after menopause.[8][9]

Pathophysiology

The androgen receptor activation shortens the anagen, or growth phase, within the normal hair growth cycle, which instigates follicular miniaturization by progressively shortening the anagen phase in androgenetic **alopecia**. Consequently, hair follicles become thinner and shorter, potentially failing not even to penetrate the epidermal layer. Pathological samples reveal a decreased 5:0 ratio of anagen to telogen hair, whereas the standard ratio is 12:1.[10][11]

Individuals with androgenetic **alopecia** exhibit elevated dihydrotestosterone (DHT) production, heightened levels of 5 alpha-reductase, and an increased abundance of androgen receptors in the areas of the scalp affected by balding. This heightened androgenic milieu in the scalp contributes to the progressive miniaturization of hair follicles, leading to the characteristic pattern of hair thinning and eventual hair loss seen in androgenetic **alopecia** (see **Image**. Androgenetic **Alopecia** in Male).

Important Background Information

Two primary isoforms of the 5 alpha-reductase enzyme exist. This enzyme converts testosterone to DHT, a compound with a higher affinity for the androgen receptor. The Type 2 isoform of the 5 alpha-reductase enzyme plays a more significant role in androgenetic **alopecia**.

The Type 1 isoform of 5 alpha-reductase enzymes are distributed within sebaceous glands, keratinocytes, and sweat glands. On the other hand, Type 2 5 alpha-reductase enzymes predominately reside in the outer root sheath of hair follicles, epididymis, vas deferens, seminal vesicles, and prostate.

The average hair growth cycle consists of 4 phases:

1. Anagen (growth)
2. Catagen (involution)
3. Telogen (resting)
4. Exogen (hair shedding)

Approximately 80% to 90% of the hair is in the anagen phase, which spans from 2 to 6 years, ultimately determining hair length. Less than 5% of the hair is in the catagen phase, while the rest are in the telogen phase. The natural exogen phase entails the shedding of around 100 hairs per day.

History and Physical

Androgenetic **alopecia** manifests as a gradual and predictable hair loss pattern after puberty. In men, it typically begins with bitemporal thinning of the frontal scalp followed by involvement of the vertex (see **Image**. Male-Pattern Androgenetic **Alopecia**). In women, the hair thinning is more diffuse, often concentrated on the crown of the scalp while retaining the frontal hairline.

The physical examination of a person with androgenetic **alopecia** involves assessing the pattern and extent of hair loss. The Norwood-Hamilton scale is commonly used to classify male-pattern baldness, while the Ludwig scale is used for female-pattern baldness. These scales help categorize the severity of hair loss and guide treatment decisions.[12]

During a physical examination, a healthcare provider might also look for signs of increased hair shedding, miniaturized hair follicles, and the overall condition of the scalp. Dermoscopy, a noninvasive technique involving a unique magnifying tool, can aid in visualizing the hair follicles and diagnosing androgenetic **alopecia**.

It's important to note that while androgenetic **alopecia** is the most common cause of hair loss, other factors such as medical conditions, medications, and stress can also contribute to hair thinning and loss. For example, in women, pattern baldness is often unmasked by telogen effluvium, which occurs 1 to 6 months after a stressor causes a larger portion of the hair to shift from the anagen phase to the telogen phase, leading to an increasing number of hair follicles in the exogen phase. Another example is that in men, early androgenetic **alopecia** is defined as occurring before 30 years and is seen as the male phenotypic equivalent to polycystic ovarian syndrome.

Evaluation

Androgenetic **alopecia** is usually diagnosed through clinical evaluation, involving a history of gradual onset after puberty and often, but not necessarily, a familial history of baldness. Biopsy is usually not necessary unless the diagnosis remains unclear. Dermoscopy can reveal miniaturized hair and brown perihilar casts, aiding in distinguishing it from diffuse **alopecia areata** that resembles male pattern baldness. Diffuse **alopecia areata** will exhibit tapered fractures such as exclamation point hairs.[8][10]

Conducting a comprehensive **review** of systems, medical history, and current medication regimen is important to rule out alternative causes for hair loss and to identify potential triggers for the manifestation of androgenetic **alopecia**. Additional studies might include thyroid studies, a complete blood count, and screening for iron deficiency through assessments of serum iron levels, total iron binding capacity, and serum ferritin levels.[13][14] If there is suspicion, screening for syphilis could also be considered.[15] Furthermore, performing a brief psychiatric assessment to identify signs of depressive symptoms and other psychiatric disorders may also be warranted for all patients seeking dermatological evaluation for hair loss.[16]

Treatment / Management

Two FDA-approved treatments are available for pattern baldness: topical minoxidil and finasteride. These treatments require at least 4 to 6 months of consistent application before noticeable improvements become apparent, and they require ongoing, indefinite usage to sustain results.[17] Due to this perpetual requirement, maintaining consistent medication adherence can be challenging. Additionally, initiation of these medications may trigger an initial shedding phase. Notably, the combined usage of these treatments often yields more favorable outcomes.[18][19][20]

Topical minoxidil is available over the counter (OTC) and is offered in various strengths, with formulations extending up to 5% solution. Higher strengths are more effective. The most common adverse effects are pruritus and local irritation, leading to flaking.[21] The flaking issue is usually due to propylene glycol or alcohol within the drug's formulation. Minoxidil functions as a potassium channel blocker, facilitating the dilation of blood vessels. This theoretical enhancement of vascular access is postulated to enable heightened oxygen, blood, and nutrient delivery to hair follicles, stimulating the anagen growth phase.

Finasteride is a 5 alpha-reductase type 2 inhibitor rather than functioning as an antiandrogen agent.[22] Typically prescribed at a daily dose of 1 mg, it displays greater effectiveness in promoting hair regrowth at the vertex region of the scalp compared to the frontal area. The efficacy of finasteride in addressing female pattern baldness remains uncertain, and it is contraindicated for women of reproductive potential (Category X) due to the risk of inducing ambiguous genitalia development in male fetuses.[23] Potential adverse effects include sexual dysfunction, which usually improves with time, an increased risk of high-grade prostate cancer due to masking of PSA levels and delayed detection, and case reports of persistent diminished libido with or without erectile dysfunction.[24][25]

Other drugs used in the management of pattern baldness lack FDA approval. Dutasteride, in particular, is noteworthy as it is approximately 3 times more potent against type II 5 alpha-reductase enzyme and 100 times more potent against the type 1 enzyme. It is often used for patients who have not experienced satisfactory results with finasteride.[26] The adverse effect profile of dutasteride is similar to that of finasteride.[27]

In women, oral antiandrogens like spironolactone are often used to treat pattern baldness. Spironolactone exhibits a very weak partial agonist toward the androgen receptor. Its primary mode of action involves blocking the more potent DHT and free testosterone from binding to the androgen receptor, physiologically behaving like a direct antagonist. Additionally, spironolactone inhibits androgen synthesis and enhances the conversion of testosterone to estradiol.

Although there haven't been documented cases of serious adverse effects associated with spironolactone use, there are some noteworthy side effects. Oral spironolactone's most commonly reported side effects include menstrual irregularities, dizziness or headache, facial hypertrichosis, rash, and hyperkalemia.[28][29] Research suggests that these side effects are usually self-resolving and manageable, with less than 5% of patients discontinuing spironolactone due to these reasons.[29]

In contrast, men using spironolactone may experience side effects like gynecomastia, decreased libido, and mild impotence.[30] These potential side effects should be discussed with a healthcare professional when considering spironolactone therapy for pattern baldness in either gender.

A similar drug, Cyproterone acetate, is used for similar purposes internationally but is not accessible in the United States.[31] Antiandrogens tend to be more effective when other signs of virilization are present, which can make them a less favorable choice to treat **alopecia** solely.[32]

Red light or laser at 660 nm has effectively addressed hair loss and is available OTC for managing pattern baldness. Low-level laser therapy has emerged as a superior nonsurgical treatment option for androgenetic **alopecia**.[33]

Hair transplantation can be effective and cosmetically satisfactory for specific patients.[34] However, patients must possess an adequate volume of donor hair, typically greater than 40 follicular units/cm^2) to adequately cover the bald area. Advances in techniques have significantly improved the cosmetic outcomes of hair transplantation, resulting in more natural and pleasing results.

Other proposed treatments for androgenetic **alopecia** include saw palmetto extract (Serenoa Repens), prostaglandin analogs like latanoprost and bimatoprost (which may be cost-prohibitive), and multiple growth factors. Platelet-rich plasma and adipose-derived stem cells are also considered in some cases, theoretically, for their potential to enhance hair growth. It is important to note that these treatments are typically not used as standalone modalities but as adjunctive therapies in combination with other established treatments for androgenetic **alopecia**.

Differential Diagnosis

Differential diagnoses of androgenetic **alopecia** encompass other hair loss disorders such as:

- **Alopecia areata**
- Anagen effluvium

- Syphilis
- Systemic diseases
- Telogen effluvium

Prognosis

Androgenetic **alopecia**, as a standalone condition, is not considered life-threatening and does not directly result in mortality. However, in some instances, this condition can be associated with other underlying severe pathologies that may coexist in a patient and potentially contribute to adverse health outcomes.

Several factors, including early onset, gender, surface area involvement, family history, treatment choices, compliance with treatment, and lifestyle factors, can influence the prognosis of androgenetic **alopecia**.[35] The prognosis for androgenetic **alopecia** is highly individualized, with outcomes varying from person to person. Although androgenetic **alopecia** is a chronic and progressive condition, early intervention and appropriate management strategies can help decrease hair loss over time, potentially leading to improved outcomes for the patient.

Complications

While some men may experience rapid progress to complete baldness within a few years, others may present with a patterned thinning of the scalp. In contrast, women often encounter hair loss at the crown of the head and rarely develop complete baldness. Once hair is lost, the scalp becomes exposed to the sun's UV rays, and evidence suggests that this exposure may heighten the risk of developing actinic keratosis.[36]

Extensive research has consistently demonstrated a strong relationship between pattern baldness, especially when it manifests early or involves vertex hair loss, and cardiovascular diseases, especially in males. These cardiovascular conditions include hypertension, myocardial infarction, insulin resistance, mortality resulting from diabetes or heart diseases, abnormal lipid profiles, obesity, and infertility.[37][38]

In females, androgenetic **alopecia** has been linked to conditions like polycystic ovarian disease. This highlights the need for comprehensive health assessments in these cases as well.

The data on the connection between androgenetic **alopecia** and benign prostate hypertrophy is inconclusive, with mixed findings. However, evidence suggests a 2-fold increased risk of prostate cancer and a higher incidence of mortality associated with prostate cancer in individuals with androgenetic **alopecia**.[39] Additionally, there is a modestly elevated risk of colon cancer and urolithiasis among patients with androgenetic **alopecia**.[40][41]

Enhancing Healthcare Team Outcomes

Androgenetic **alopecia** is a prevalent condition that can significantly impact the appearance and self-esteem of both males and females. **Alopecia** is a common issue encountered in dermatology clinics, yet many patients feel embarrassed to seek medical assistance, often resorting to online solutions. Essential to addressing this disorder is patient education.

Treating androgenetic **alopecia** often involves an interprofessional team of healthcare practitioners working together to provide comprehensive care. An interprofessional team can collaborate to treat androgenetic **alopecia** in the following ways:

- **Dermatologist**: The primary healthcare provider responsible for diagnosing and managing androgenetic **alopecia**. They can perform a thorough evaluation, recommend appropriate treatments, and monitor the condition's progress.

- **Primary care physician**: PCPs, including physicians and advanced practitioners, can be critical in identifying risk factors for androgenetic **alopecia** during routine check-ups. They may refer patients to dermatologists for specialized care and coordinate overall healthcare.
- **Pharmacist**: The pharmacist is valuable in educating patients about medications used to treat androgenetic **alopecia**, such as minoxidil and finasteride. They can provide information on proper usage, potential side effects, and drug interactions, ensuring patients adhere to their treatment plans.
- **Endocrinologist**: In cases where hormonal imbalances are suspected, an endocrinologist may be consulted to assess and manage hormonal issues contributing to androgenetic **alopecia**.
- **Plastic surgeon**: Patients may consider hair transplant surgery in advanced cases of androgenetic **alopecia** where medical treatments have not yielded satisfactory results. Plastic surgeons can perform these procedures.
- **Nutritionist/Dietitian**: Dietary factors can influence hair health. Nutritionists or dietitians can guide dietary changes and nutritional supplements that may promote hair growth and overall hair health.
- **Psychologist**: Hair loss can have a significant psychological impact on individuals. Psychologists or counselors can help patients cope with the emotional and psychological effects of hair loss, offering strategies to improve self-esteem and body image.

Despite the availability of various medications and hair transplant procedures, the outcomes of treatments for androgenetic **alopecia** remain unpredictable. Providing simple reassurance that this is a benign disorder may suffice for patients not significantly distressed by hair thinning. Clinicians and pharmacists must collaborate in educating patients about this emotionally challenging issue, ensuring they are well-informed about available options, potential risks, and expected outcomes.[4][42][43]

Review Questions

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Figures



Figure 1

Male-Pattern Androgenetic **Alopecia**. Photo of typical hair loss in males with androgenetic **alopecia**. Contributed by A Al Aboud, MD



Figure 2

Male-Pattern Androgenetic **Alopecia**. Hair loss is seen in the frontal, temporal, and central scalp areas. Contributed by A Al Aboud, MD

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